

Higher-Dimensional Differential Geometry and Topology

Akul Kumar

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Abstract

In this paper, we provide an organized introduction to the study of higher-dimensional differential geometry and topology. We begin by going over the fundamentals of vector fields, tangent and cotangent structures, smooth manifolds, and Riemannian metrics. Next, we demonstrate how curvature concepts can be extended to higher dimensions and how the Gauss-Bonnet-Chern theorem relates them to topology. We then provide an introduction to cobordism and surgery theory in higher dimensions, after which Morse theory is developed as a crucial method for examining manifold topology. Characteristic classes (including the Euler, Chern, and Pontryagin classes) and their function in connecting local and global structure will also be covered. The existence of infinitely many smooth structures on $\mathbb{R}^4$ is one of the many reasons why manifolds of dimension $n \geq 4$ lead to phenomena which do not occur in lower dimensions. There also exist many connections related to physics; for one, the configuration spaces in general relativity, gauge theory, and string theory are inherently higher-dimensional manifolds. In addition, higher-dimensional manifolds need complex invariants like characteristic classes and cobordism groups, whereas surfaces are categorized by genus. So, as a concluding note, we also provide a brief overview of applications to physics and mathematics, such as the Atiyah-Singer Index Theorem.

Contents

1	Introduction	2
2	Smooth Manifolds, Tangent and Cotangent Structures	3
2.1	Topological Groundwork	3
2.2	Tangent Spaces and Bundles	3
2.3	Cotangent Spaces and Differential Forms	4
2.4	Riemannian Metrics	5
3	Higher-Dimensional Curvature	5
3.1	The Levi-Civita Connection	5
3.2	The Riemann Curvature Tensor	6
3.3	Sectional, Ricci, and Scalar Curvature	6
3.4	The Chern-Gauss-Bonnet Theorem	6

4	Morse Theory	7
4.1	Morse Functions and Critical Points	7
4.2	The Morse Lemma	7
4.3	Sublevel Sets and Handle Attachments	7
4.4	Morse Inequalities	8
4.5	Examples	8
5	Characteristic Classes	8
5.1	Vector Bundles and Connections	8
5.2	Chern-Weil Theory	9
5.3	Chern, Pontryagin, and Euler Classes	9
5.4	Examples	9
5.5	Relation to Topological Invariants	10
6	Cobordism, h Cobordism, and Surgery Theory	10
6.1	Cobordism and h-cobordism	10
6.2	Handles and cancellation	10
6.3	The h-cobordism and s-cobordism theorems	11
6.4	Surgery on manifolds	11
6.5	Classification applications	11
6.6	Wall's surgery exact sequence	12
6.7	Worked examples	12
6.8	Summary	12
7	Applications: The Atiyah-Singer Index Theorem	13
7.1	Elliptic differential operators	13
7.2	Analytical and topological index	13
7.3	The theorem	13
7.4	Special cases	14
7.5	Geometric significance	14
7.6	Summary	14

1 Introduction

Classical results from curves and surfaces are extended to smooth manifolds of arbitrary dimension in higher dimensional Differential Geometry and Topology. In dimensions $n \geq 3$, new and unexpected phenomena appear, like exotic smooth structures on spheres. Certain connections observed in low dimensions, like the Gauss–Bonnet connection between integrated curvature and topology in 2 dimensions, do not correspondingly generalize in higher dimensions. The fact that higher-dimensional manifolds display a wider range of topological types and geometric behaviors means that tools from geometric analysis (Riemannian metrics, curvature forms) and algebraic topology (homology, cohomology, characteristic classes) are required.

Foundational ideas for this topic were developed in the 19th century; primarily by Gauss, Riemann, and Poincaré. By the mid 20th century, Smale, Thom, Milnor, Chern, among others had made important contributions to higher dimensional differential topology. Important results such as the Chern-Gauss-Bonnet theorem, the h -cobordism theorem, and the Atiyah-Singer index theorem came about in this time period.

2 Smooth Manifolds, Tangent and Cotangent Structures

The basic objects of differential geometry (smooth manifolds, their tangent and cotangent spaces, and the corresponding tensor fields) are developed in this section.

2.1 Topological Groundwork

Definition 2.1. A **topological manifold** of dimension n is a Hausdorff, second countable, paracompact topological space M such that each point $p \in M$ has a neighborhood homeomorphic to an open subset of $\mathbb{R}^n$. The local homeomorphisms are called **charts**.

Definition 2.2. Let M be a n dimensional topological manifold. A pair (U, φ) is called a **chart** if $U \subset M$ is open and $\varphi : U \rightarrow \varphi(U) \subset \mathbb{R}^n$ is a homeomorphism onto an open set. A collection $\mathcal{A} = \{(U_\alpha, \varphi_\alpha)\}_{\alpha \in A}$ of charts whose domains cover M is called an **atlas**.

A **transition map**, given two overlapping charts (U, φ) and (V, ψ) , $\psi \circ \varphi^{-1} : \varphi(U \cap V) \rightarrow \psi(U \cap V)$, encodes how coordinates change. The smoothness of the transition map determines how smooth a manifold is.

Definition 2.3. A **smooth atlas** is one in which every transition map $\psi \circ \varphi^{-1}$ is C^∞ . A topological manifold with a maximally smooth atlas is called a **smooth manifold**.

Remark 2.4. A unique maximal smooth atlas, which is created by incorporating all charts that are smoothly in line with the specified one, contains all smooth atlases.

Example 2.5. The n -sphere $S^n = \{x \in \mathbb{R}^{n+1} \mid \|x\| = 1\}$ given atlas which is also given by stereographic projection from the north and south poles.

2.2 Tangent Spaces and Bundles

There are several formal definitions for the tangent space $T_p M$ of a smooth manifold M at $p \in M$. The definition given below is the derivation definition.

Definition 2.6. The **tangent space** $T_p M$ is the $\mathbb{R}$ -vector space of derivations at p , i.e.,

$$T_p M := \{X_p : C^\infty(M) \rightarrow \mathbb{R} \mid X_p \text{ is linear and } X_p(fg) = X_p(f)g(p) + f(p)X_p(g)\}.$$

Given a chart (U, φ) with local coordinates $(x^1, \dots, x^n)$, for each i we define the derivation

$$\left. \frac{\partial}{\partial x^i} \right|_p (f) := \left. \frac{\partial(f \circ \varphi^{-1})}{\partial u^i} \right|_{\varphi(p)}, \quad f \in C^\infty(M).$$

These are elements of $T_p M$ called the *coordinate vector fields* at p .

Theorem 2.7. *If (U, φ) is a chart with coordinates $(x^1, \dots, x^n)$ and $p \in U$, then $\{\partial/\partial x^i|_p\}_{i=1}^n$ is a basis for $T_p M$.*

Proof. Given $f \in C^\infty(M)$, the coordinate function x^i induces partial derivatives $\frac{\partial}{\partial x^i}|_p f = \frac{\partial(f \circ \varphi^{-1})}{\partial u^i}|_{\varphi(p)}$. These satisfy the derivation property, and any derivation X_p is determined by its action on the coordinate functions, establishing the basis property. $\square$

Definition 2.8. *The **tangent bundle** of a smooth n -manifold M is the disjoint union*

$$TM := \bigsqcup_{p \in M} T_p M,$$

together with the projection $\pi : TM \rightarrow M$ sending $v \in T_p M$ to p . To make TM into a smooth manifold of dimension $2n$, proceed as follows: for each chart (U, φ) with coordinates $(x^1, \dots, x^n)$, define a map

$$\Psi_U : \pi^{-1}(U) \longrightarrow U \times \mathbb{R}^n, \quad \Psi_U \left(v_1 \frac{\partial}{\partial x^1} \Big|_p + \dots + v_n \frac{\partial}{\partial x^n} \Big|_p \right) = (p, (v_1, \dots, v_n)).$$

The collection $\{\Psi_U\}$ forms an atlas making TM into a smooth manifold, and each fiber $\pi^{-1}(p) = T_p M$ is a vector space. With this structure, (TM, π, M) is a smooth rank- n vector bundle, called the tangent bundle.

2.3 Cotangent Spaces and Differential Forms

The cotangent space $T_p^* M$ is the dual space of $T_p M$. Elements of these are referred to as **covectors**. The dual basis to $\{\partial/\partial x^i|_p\}$ is denoted as $\{dx^i|_p\}$.

Definition 2.9 (Differential k -forms). *Let M be a smooth n -manifold. For each $p \in M$, the exterior power $\Lambda^k T_p^* M$ is the vector space of alternating k -linear forms on $T_p M$. The disjoint union*

$$\Lambda^k T^* M := \bigsqcup_{p \in M} \Lambda^k T_p^* M$$

*forms a smooth vector bundle over M , called the **bundle of k -covectors**. A **differential k -form** on M is a smooth section of this bundle.*

The space of all differential k -forms is denoted

$$\Omega^k(M) := \Gamma(\Lambda^k T^* M).$$

Locally, any $\omega \in \Omega^k(M)$ can be written as

$$\omega = \sum_{1 \leq i_1 < \dots < i_k \leq n} f_{i_1 \dots i_k} dx^{i_1} \wedge \dots \wedge dx^{i_k},$$

where the $f_{i_1 \dots i_k} \in C^\infty(U)$ are smooth coefficient functions.

Definition 2.10. *The unique $\mathbb{R}$ -linear map satisfying:*

1. $d^2 = 0$,
2. $d(f) = df$ is the usual differential for $f \in C^\infty(M)$,
3. $d(\alpha \wedge \beta) = d\alpha \wedge \beta + (-1)^k \alpha \wedge d\beta$ for $\alpha \in \Omega^k(M)$.

is known as the **exterior derivative** $d : \Omega^k(M) \rightarrow \Omega^{k+1}(M)$

Theorem 2.11 (Poincaré Lemma). *If $U \subset M$ is contractible, then every closed form $\omega \in \Omega^k(U)$ ($d\omega = 0$) is exact: $\omega = d\eta$ for some $\eta \in \Omega^{k-1}(U)$.*

2.4 Riemannian Metrics

A smooth section $g \in \Gamma(\text{Sym}^2 T^*M)$ such that g_p is a positive-definite inner product on $T_p M$ for all $p \in M$ is known as the **Riemannian metric** on M . The pair (M, g) is a **Riemannian manifold**.

Example 2.12. *The round metric on $S^n \subset \mathbb{R}^{n+1}$ is the restriction of the Euclidean metric δ_{ij} to the tangent spaces of S^n .*

3 Higher-Dimensional Curvature

In a Riemannian manifold (M, g) , curvature is the deviation of the geometry from being locally Euclidean. Curvature is defined by a $(1, 3)$ tensor in higher dimensions. This tensor is known as the **Riemann curvature tensor**.

3.1 The Levi-Civita Connection

A connection on TM is a map

$$\nabla : \mathfrak{X}(M) \times \mathfrak{X}(M) \rightarrow \mathfrak{X}(M), \quad (X, Y) \mapsto \nabla_X Y,$$

satisfying:

1. $\nabla_{fX+gY} Z = f\nabla_X Z + g\nabla_Y Z$,
2. $\nabla_X (Y + Z) = \nabla_X Y + \nabla_X Z$,
3. $\nabla_X (fY) = X(f)Y + f\nabla_X Y$,

for $X, Y, Z \in \mathfrak{X}(M)$, $f, g \in C^\infty(M)$.

Theorem 3.1 (Fundamental Theorem of Riemannian Geometry). *There exists a unique connection ∇ on TM for any Riemannian manifold (M, g) :*

1. **Metric-compatible:** $X(g(Y, Z)) = g(\nabla_X Y, Z) + g(Y, \nabla_X Z)$.
2. **Torsion-free:** $\nabla_X Y - \nabla_Y X = [X, Y]$.

*This connection is known as the **Levi-Civita connection**.*

Proof. The **Koszul formula** introduces existence and uniqueness:

$$2g(\nabla_X Y, Z) = Xg(Y, Z) + Yg(Z, X) - Zg(X, Y) - g(X, [Y, Z]) + g(Y, [Z, X]) + g(Z, [X, Y]),$$

Thus, $\nabla_X Y$ is uniquely determined from g and the Lie bracket. $\square$

3.2 The Riemann Curvature Tensor

Definition 3.2. For a (M, g) with Levi-Civita connection ∇ , the **Riemann curvature tensor** is the $(1, 3)$ -tensor

$$R(X, Y)Z := \nabla_X \nabla_Y Z - \nabla_Y \nabla_X Z - \nabla_{[X, Y]} Z.$$

In local coordinates $\{x^i\}$ with Christoffel symbols Γ_{ij}^k ,

$$R^l_{ijk} = \partial_j \Gamma_{ik}^l - \partial_i \Gamma_{jk}^l + \Gamma_{ik}^m \Gamma_{jm}^l - \Gamma_{jk}^m \Gamma_{im}^l.$$

Theorem 3.3 (Symmetries of R). For all $X, Y, Z, W \in \mathfrak{X}(M)$:

1. $R(X, Y) = -R(Y, X)$.
2. $g(R(X, Y)Z, W) = -g(R(X, Y)W, Z)$.
3. **First Bianchi identity:** $R(X, Y)Z + R(Y, Z)X + R(Z, X)Y = 0$.
4. **Second Bianchi identity:** $\nabla_X R(Y, Z) + \nabla_Y R(Z, X) + \nabla_Z R(X, Y) = 0$.

3.3 Sectional, Ricci, and Scalar Curvature

Definition 3.4. Let $p \in M$ and $\sigma \subset T_p M$ be a 2-dimensional subspace with orthonormal basis $\{u, v\}$. The **sectional curvature** of σ is

$$K(\sigma) = g(R(u, v)v, u).$$

Definition 3.5. The $(0, 2)$ -tensor below is known as the **Ricci curvature**:

$$\text{Ric}(X, Y) = \text{trace}\{Z \mapsto R(Z, X)Y\}.$$

$S = \text{trace}_g \text{Ric}$ is known as the **scalar curvature**.

Example 3.6. For the n -sphere (S^n, g_{round}) of radius r , $K(\sigma) \equiv \frac{1}{r^2}$ for all σ ; hence $\text{Ric} = \frac{n-1}{r^2}g$ and $S = \frac{n(n-1)}{r^2}$.

3.4 The Chern-Gauss-Bonnet Theorem

The Gauss-Bonnet theorem relates the total Gaussian curvature to the Euler characteristic in two dimensions:

$$\int_M K dA = 2\pi \chi(M).$$

The **Pfaffian** of the curvature form is used for higher even dimensional curvatures.

Theorem 3.7 (Chern-Gauss-Bonnet). Let (M^{2n}, g) be a closed oriented Riemannian manifold. Let Ω be the curvature 2-form of the Levi-Civita connection. Then

$$\chi(M) = \frac{1}{(2\pi)^n} \int_M \text{Pf}(\Omega),$$

where $\text{Pf}(\Omega)$ is the Pfaffian, a $2n$ -form representing the Euler class in de Rham cohomology.

4 Morse Theory

The critical points of real-valued smooth functions defined on a smooth manifold are related to its differential topology by Morse theory. Specifically, it offers an analytical framework for handling decompositions and explains how the topology changes as it passes through critical values.

4.1 Morse Functions and Critical Points

Definition 4.1. Let M be a smooth manifold and $f \in C^\infty(M)$. A point $p \in M$ is a **critical point** of f if $df_p = 0$. The **Hessian** of f at p is the symmetric bilinear form

$$\text{Hess}_p(f)(X, Y) := X(Yf) - (\nabla_X Y)f$$

on $T_p M$, where ∇ is the Levi-Civita connection of some Riemannian metric on M . We say p is **non-degenerate** if $\text{Hess}_p(f)$ is non-singular.

Definition 4.2. A smooth function $f : M \rightarrow \mathbb{R}$ is a **Morse function** if all its critical points are non-degenerate.

Remark 4.3. Since non-degeneracy is a coordinate-invariant property of the second derivative at p , the definition is independent of the choice of metric.

Example 4.4. The height function $f : S^2 \rightarrow \mathbb{R}$ given by $f(x, y, z) = z$ is Morse, with exactly two critical points: the north and south poles.

4.2 The Morse Lemma

Lemma 4.5 (Morse Lemma). Let $f : M \rightarrow \mathbb{R}$ be a smooth function and $p \in M$ a non-degenerate critical point of index λ (i.e., $\text{Hess}_p(f)$ has λ negative eigenvalues). Then there exist local coordinates $(x^1, \dots, x^n)$ centered at p such that

$$f(x) = f(p) - (x^1)^2 - \dots - (x^\lambda)^2 + (x^{\lambda+1})^2 + \dots + (x^n)^2.$$

Proof. Choose coordinates such that $g_{ij}(p) = \delta_{ij}$ and the Hessian is diagonal at p by the spectral theorem. The Taylor expansion of f at p has no linear terms and quadratic terms matching the Hessian. Higher-order terms can be eliminated by an explicit smooth change of variables, yielding the claimed normal form. $\square$

4.3 Sublevel Sets and Handle Attachments

For $a \in \mathbb{R}$, define the **sublevel set** $M_a := f^{-1}((-\infty, a])$. If $[a, b]$ contains no critical values, M_a and M_b are diffeomorphic. The topology changes precisely when passing a critical value.

Theorem 4.6 (Handle Attachment Theorem). Let $f : M \rightarrow \mathbb{R}$ be a Morse function and suppose c is a critical value with exactly one critical point p of index λ . Then for $\varepsilon > 0$ small, $M_{c+\varepsilon}$ is diffeomorphic to $M_{c-\varepsilon}$ with a λ -handle $D^\lambda \times D^{n-\lambda}$ attached along $\partial D^\lambda \times D^{n-\lambda}$.

4.4 Morse Inequalities

Definition 4.7. Let m_λ denote the number of index- λ critical points of f . Let $b_\lambda = \dim H_\lambda(M; \mathbb{Q})$ be the λ -th Betti number.

Theorem 4.8 (Weak Morse Inequalities). For each λ ,

$$m_\lambda \geq b_\lambda.$$

Theorem 4.9 (Strong Morse Inequalities). For each k ,

$$\sum_{\lambda=0}^k (-1)^{k-\lambda} m_\lambda \geq \sum_{\lambda=0}^k (-1)^{k-\lambda} b_\lambda,$$

with equality for $k = n$:

$$\sum_{\lambda=0}^n (-1)^\lambda m_\lambda = \sum_{\lambda=0}^n (-1)^\lambda b_\lambda = \chi(M).$$

4.5 Examples

Example 4.10 (Morse Function on S^n). Let $f : S^n \rightarrow \mathbb{R}$ be the height function in $\mathbb{R}^{n+1}$. It has two critical points: the north pole (index n) and the south pole (index 0). The Morse inequalities are sharp: $m_0 = b_0 = 1$, $m_n = b_n = 1$, all others vanish.

Example 4.11 (Morse Function on T^2). Let $T^2 = S^1 \times S^1$ embedded in $\mathbb{R}^3$, and take f to be the vertical projection. One finds $m_0 = 1$, $m_1 = 2$, $m_2 = 1$, matching the Betti numbers $b_0 = 1$, $b_1 = 2$, $b_2 = 1$.

Remark 4.12. Morse-Bott functions are natural extensions of Morse theory. Here, critical points form non-degenerate submanifolds; leading to richer topology changes.

5 Characteristic Classes

Topological invariants associated to vector bundles are known as **Characteristic Classes**.

5.1 Vector Bundles and Connections

Definition 5.1. A **smooth real vector bundle** of rank k over a smooth manifold M is a smooth manifold E together with a surjective smooth map $\pi : E \rightarrow M$ such that for each $p \in M$ there exists an open neighborhood U and a diffeomorphism

$$\Phi : \pi^{-1}(U) \rightarrow U \times \mathbb{R}^k$$

making the diagram commute and restricting to linear isomorphisms on fibers.

Definition 5.2. A **connection** ∇ on a vector bundle $E \rightarrow M$ is a map

$$\nabla : \mathfrak{X}(M) \times \Gamma(E) \rightarrow \Gamma(E)$$

satisfying:

$$1. \nabla_{fX+gY}s = f\nabla_X s + g\nabla_Y s,$$

$$2. \nabla_X(fs) = X(f)s + f\nabla_X s,$$

for $X, Y \in \mathfrak{X}(M)$, $f, g \in C^\infty(M)$, $s \in \Gamma(E)$.

Definition 5.3. The curvature of ∇ is the bundle endomorphism-valued 2-form

$$\Omega(X, Y) := \nabla_X \nabla_Y - \nabla_Y \nabla_X - \nabla_{[X, Y]}.$$

5.2 Chern-Weil Theory

Let E be a rank- k complex vector bundle with connection ∇ and curvature $\Omega \in \Omega^2(M; \mathfrak{gl}_k(\mathbb{C}))$.

Theorem 5.4 (Chern-Weil Homomorphism). *Let P be an invariant polynomial of degree r on $\mathfrak{gl}_k(\mathbb{C})$. Then $P\left(\frac{i}{2\pi}\Omega\right)$ is a closed $2r$ -form on M , and its de Rham cohomology class is independent of the choice of connection ∇ . This defines a characteristic class of E .*

5.3 Chern, Pontryagin, and Euler Classes

Definition 5.5. For a complex vector bundle E ,

$$c(E) = \det\left(I + \frac{i}{2\pi}\Omega\right) = 1 + c_1(E) + c_2(E) + \cdots + c_k(E),$$

where $c_j(E) \in H^{2j}(M; \mathbb{Z})$ are the **Chern classes**.

Definition 5.6. For a real vector bundle E , the Pontryagin classes are defined via the complexification $E_{\mathbb{C}}$:

$$p_j(E) = (-1)^j c_{2j}(E_{\mathbb{C}}) \in H^{4j}(M; \mathbb{Z}).$$

Definition 5.7. If E is oriented of rank n , the **Euler class** $e(E) \in H^n(M; \mathbb{Z})$ is the top Chern class of the complexified bundle when n is even, or equivalently the cohomology class represented by the Pfaffian of $\frac{1}{2\pi}\Omega$.

5.4 Examples

Example 5.8 (S^2). Let TS^2 be the tangent bundle of the round 2-sphere of radius r . The Gaussian curvature $K = \frac{1}{r^2}$ is constant. The Euler class $e(TS^2)$ is represented by $\frac{K}{2\pi} dA$, whose integral is

$$\frac{1}{2\pi} \int_{S^2} \frac{1}{r^2} r^2 \sin \theta d\theta d\phi = 2,$$

matching $\chi(S^2) = 2$.

Example 5.9 ($\mathbb{CP}^n$). Let $\mathcal{O}(1)$ denote the hyperplane bundle over $\mathbb{CP}^n$. The Fubini-Study metric yields curvature Ω whose cohomology class represents $c_1(\mathcal{O}(1)) = \alpha \in H^2(\mathbb{CP}^n; \mathbb{Z})$. Then $c(\mathcal{O}(1)) = 1 + \alpha$ and the total Chern class of $T\mathbb{CP}^n$ is $(1 + \alpha)^{n+1}$.

5.5 Relation to Topological Invariants

Theorem 5.10. *If M is a closed oriented $2n$ -manifold, then*

$$\langle e(TM), [M] \rangle = \chi(M).$$

Proof. This is the generalized Gauss-Bonnet theorem: $e(TM)$ is represented by the Pfaffian of the curvature form, whose integral equals $\chi(M)$. $\square$

Remark 5.11. *Characteristic classes obstruct certain geometric structures: e.g., if $e(TM) \neq 0$, M admits no nowhere-vanishing vector field.*

6 Cobordism, hCobordism, and Surgery Theory

Cobordism and surgery provide a high-dimensional calculus for modifying manifolds while controlling their homotopy type and characteristic invariants. In dimensions ≥ 5 , these methods yield powerful classification theorems (Smale, Barden, Kervaire-Milnor, Wall) that contrast sharply with the wildness of dimension 4.

6.1 Cobordism and h-cobordism

Definition 6.1. *Two closed n -manifolds M_0, M_1 are **cobordant** if there exists a compact $(n+1)$ -manifold W with $\partial W = M_0 \sqcup M_1$. We write $M_0 \sim M_1$. The set of cobordism classes under disjoint union forms an abelian group Ω_n^O (oriented: Ω_n^{SO}).*

Definition 6.2. *An $(n+1)$ -dimensional cobordism $(W; M_0, M_1)$ is an **h-cobordism** if the inclusions $M_i \hookrightarrow W$ are homotopy equivalences. It is an **s-cobordism** if, in addition, the simple homotopy classes are trivial; equivalently, the Whitehead torsion $\tau(M_i \hookrightarrow W) \in \text{Wh}(\pi_1 M_i)$ vanishes.*

Theorem 6.3 (Poincaré-Lefschetz Duality for Cobordisms). *Let $(W; M_0, M_1)$ be a cobordism. Then there is a long exact sequence relating the pairs (W, M_0) and (W, M_1) and Poincaré-Lefschetz duality exchanges relative and absolute (co)homology. In particular, if $(W; M_0, M_1)$ is an h-cobordism, $H_*(W, M_i) = 0$ for $i = 0, 1$.*

6.2 Handles and cancellation

Every compact cobordism admits a handle decomposition relative to one boundary.

Theorem 6.4 (Relative Handle Decomposition). *Let $(W; M_0, M_1)$ be a cobordism. Then W is obtained from $M_0 \times [0, 1]$ by attaching handles of indices $1, \dots, n$ along $M_0 \times \{1\}$. If $(W; M_0, M_1)$ is an h-cobordism, the handle decomposition can be chosen without 0- or $(n+1)$ -handles, and the relative homology vanishing forces algebraic cancellations between indices k and $k+1$.*

Lemma 6.5 (Handle Cancellation Criterion). *Suppose W has a k -handle $h^k = D^k \times D^{n+1-k}$ whose attaching sphere $S^{k-1} \subset \partial_+(M_0 \times [0, 1] \cup \text{lower handles})$ intersects the belt sphere of some $(k-1)$ -handle transversely in a single point. Then the pair cancels: removing both handles yields a diffeomorphic cobordism.*

6.3 The h-cobordism and s-cobordism theorems

Theorem 6.6 (Smale's h-Cobordism Theorem). *Let $(W; M_0, M_1)$ be a smooth h-cobordism of dimension $n + 1 \geq 6$ (i.e. $n \geq 5$) with M_0, M_1 closed and simply connected. Then $W \cong M_0 \times [0, 1]$ (diffeomorphic rel boundary). In particular $M_0 \cong M_1$.*

Theorem 6.7 (s-Cobordism Theorem). *Let $(W; M_0, M_1)$ be an h-cobordism of dimension $n + 1 \geq 6$. Then $W \cong M_0 \times [0, 1]$ rel boundary if and only if the Whitehead torsion $\tau(M_0 \hookrightarrow W) \in \text{Wh}(\pi_1 M_0)$ vanishes. Moreover, the assignment $W \mapsto \tau$ induces a bijection between isomorphism classes of h-cobordisms (rel M_0) and $\text{Wh}(\pi_1 M_0)$.*

Remark 6.8. *For simply connected M_0 , $\text{Wh}(1) = 0$, so h-cobordism implies product (Smale). In general, $\text{Wh}(\pi_1)$ governs the obstruction to cancelling all handles.*

6.4 Surgery on manifolds

Definition 6.9. *Let M^n be a smooth manifold and $S^p \subset M$ an embedded sphere with trivial normal bundle ($\nu \cong S^p \times D^{q+1}$, $p + q + 1 = n$). Remove a tubular neighborhood and glue back $D^{p+1} \times S^q$ along $S^p \times S^q = \partial(S^p \times D^{q+1}) = \partial(D^{p+1} \times S^q)$. The resulting manifold M' is obtained by a (p, q) -surgery on M .*

Lemma 6.10 (Effect on homotopy). *If the attaching map kills a chosen element of $\pi_p(M)$ and $q \geq 2$, then surgery in index p modifies π_* only in controlled degrees (it kills that element and leaves π_i unchanged for $i < p$, up to natural basepoint issues).*

Theorem 6.11 (Surgery to a homotopy equivalence). *Let $f : M \rightarrow X$ be a degree-one normal map between closed n -manifolds with compatible stable normal bundles. If the associated surgery obstruction $\theta(f) \in L_n(\mathbb{Z}[\pi_1 X])$ vanishes (Wall group), then one can perform a finite sequence of surgeries on M (in indices $\leq \lfloor n/2 \rfloor$) to obtain M' and a simple homotopy equivalence $f' : M' \simeq X$.*

6.5 Classification applications

Exotic spheres (Kervaire-Milnor). Let Θ_n denote the group of h-cobordism classes of homotopy n -spheres under connected sum.

Theorem 6.12 (Kervaire-Milnor). *For $n \geq 5$, Θ_n is finite. In particular, $\Theta_7 \cong \mathbb{Z}/28$. Hence there are 28 distinct smooth structures on the topological 7-sphere.*

Idea. Identify homotopy spheres as boundaries of parallelizable manifolds and analyze the stable framing obstructions via the J -homomorphism and surgery exact sequence. The finiteness follows from the finiteness of relevant stable homotopy groups and L -groups. $\square$

Simply connected 5-manifolds (Barden).

Theorem 6.13 (Barden classification). *Closed, simply connected, smooth 5-manifolds are classified up to diffeomorphism by their second homology (as an abelian group with linking form on torsion), the second Stiefel-Whitney class, and certain additional invariants.*

High-dimensional Poincaré conjecture (Smale).

Theorem 6.14 (Generalized Poincaré in $n \geq 5$). *A closed simply connected homology n -sphere ($n \geq 5$) is homeomorphic (indeed, diffeomorphic except in dimensions with exotic spheres) to S^n .*

Remark 6.15 (Dimension 4). *Surgery fails in general in dimension 4 due to the failure of the Whitney trick. Freedman proved the topological classification of simply connected closed 4-manifolds, while Donaldson showed strong smooth obstructions; hence the sharp split between topological and smooth categories in dimension 4.*

6.6 Wall's surgery exact sequence

For a finite CW complex X of formal dimension n and a normal invariant $[M, f, \nu] \in \mathcal{N}(X)$ (degree-one normal map), surgery theory fits into the exact sequence

$$\cdots \longrightarrow \mathcal{S}(X) \longrightarrow \mathcal{N}(X) \xrightarrow{\theta} L_n(\mathbb{Z}[\pi_1 X]) \xrightarrow{\partial} \mathcal{S}_{n-1}(X) \longrightarrow \cdots$$

where $\mathcal{S}(X)$ is the structure set (simple homotopy types of manifolds homotopy equivalent to X) and L_n denotes Wall's L -groups.

Theorem 6.16 (Consequences). *If $L_n(\mathbb{Z}[\pi_1 X]) = 0$ and the normal invariant lifts, then every degree-one normal map is normally bordant to a simple homotopy equivalence; thus the homotopy type X determines the manifold up to s -cobordism.*

6.7 Worked examples

Example 6.17 (Killing π_1 by 1-surgery in $n \geq 5$). *Let M^n ($n \geq 5$) have finitely generated $\pi_1(M)$. Choose disjoint embedded loops representing generators with trivial normal bundles. Performing 1-surgeries kills these generators, yielding a new manifold M' with simplified π_1 . Subsequent higher-index surgeries can adjust higher homotopy groups toward a target simply connected model, provided L -theory obstructions vanish.*

Example 6.18 (Surgery on $S^p \times S^q$). *For $n = p + q$, $S^p \times S^q$ admits a (p, q) -surgery along $S^p \times \{*\}$ producing the connected sum $S^{p+1} \times S^{q-1} \# (\text{possible exotic pieces})$ depending on framings. Iterating such surgeries under obstruction control leads to classification results in certain dimensions.*

6.8 Summary

In dimensions ≥ 5 , h-cobordism and surgery theory reduce manifold classification to algebraic invariants (simple homotopy type, characteristic classes, and Wall's L -groups). This yields finiteness of exotic spheres, classification of many simply connected manifolds, and the high-dimensional Poincaré conjecture. The failure of the Whitney trick in dimension 4 delineates the boundary of these methods' effectiveness.

7 Applications: The Atiyah-Singer Index Theorem

One of the deepest applications of higher-dimensional differential geometry and topology is the **Atiyah-Singer Index Theorem**, which connects analysis, geometry, and topology by expressing the analytical index of an elliptic differential operator in terms of topological data derived from characteristic classes.

7.1 Elliptic differential operators

Let M be a closed smooth manifold and $E, F \rightarrow M$ be complex vector bundles.

Definition 7.1. A **differential operator** $D : \Gamma(E) \rightarrow \Gamma(F)$ of order m is a $\mathbb{C}$ -linear map which in local coordinates $(x^1, \dots, x^n)$ can be written as

$$D = \sum_{|\alpha| \leq m} a_\alpha(x) \partial^\alpha$$

with smooth coefficient matrices $a_\alpha(x)$.

Definition 7.2. The **symbol** of D is the bundle map

$$\sigma_D : \pi^* E \longrightarrow \pi^* F, \quad \sigma_D(x, \xi) = \sum_{|\alpha|=m} a_\alpha(x) \xi^\alpha,$$

defined on the cotangent bundle T^*M .

Definition 7.3. D is **elliptic** if $\sigma_D(x, \xi)$ is an isomorphism for all $(x, \xi) \in T^*M$ with $\xi \neq 0$.

Ellipticity ensures that the solutions of $Ds = 0$ are finite-dimensional and that D has a well-defined Fredholm index.

7.2 Analytical and topological index

Definition 7.4. For an elliptic operator D , its **analytical index** is

$$\text{ind}_a(D) := \dim \ker D - \dim \text{coker } D.$$

Definition 7.5. The **topological index** $\text{ind}_t(D)$ is computed from the symbol class $[\sigma_D] \in K^0(T^*M)$ via the Thom isomorphism and the pushforward map in K -theory, expressed as an integral over M of characteristic classes:

$$\text{ind}_t(D) = \langle \text{ch}(\sigma_D) \smile \text{Td}(TM \otimes \mathbb{C}), [M] \rangle.$$

Here ch is the Chern character and Td is the Todd class.

7.3 The theorem

Theorem 7.6 (Atiyah-Singer Index Theorem). *Let M be a closed smooth manifold and D an elliptic differential operator. Then*

$$\text{ind}_a(D) = \text{ind}_t(D).$$

In particular, the analytical index (an analytic quantity) equals the topological index (computed from characteristic classes).

7.4 Special cases

Example 7.7 (Gauss-Bonnet as an index theorem). *Let M^{2n} be a closed oriented Riemannian manifold, and let $D = d + d^*$ be the de Rham operator from even to odd forms. D is elliptic, and $\ker D$ corresponds to harmonic forms. Hodge theory identifies the analytical index with*

$$\text{ind}_a(D) = \sum_{k=0}^n (-1)^k b_k(M) = \chi(M).$$

The topological index $\text{ind}_t(D)$ equals $\int_M e(TM)$ via the Chern-Weil description, yielding the Chern-Gauss-Bonnet theorem as a corollary.

Example 7.8 (Riemann-Roch theorem). *For a compact complex manifold X and a holomorphic vector bundle E , the Dolbeault operator $\bar{\partial}_E$ is elliptic. The index theorem specializes to the Hirzebruch-Riemann-Roch formula:*

$$\chi(X, E) = \int_X \text{ch}(E) \smile \text{Td}(TX).$$

Example 7.9 (Signature theorem). *Let M^{4k} be an oriented $4k$ -dimensional manifold. The signature operator D has analytical index equal to the signature of the intersection form on $H^{2k}(M; \mathbb{R})$. The index theorem yields Hirzebruch's formula:*

$$\sigma(M) = \int_M L(p_1(TM), \dots, p_k(TM)),$$

where L is the L -polynomial in Pontryagin classes.

7.5 Geometric significance

As demonstrated by the index theorem, some purely analytical quantities are actually topological invariants. This is a profound fact; curvature forms are produced by the choice of metric and connection, which allows geometry to enter. The characteristic classes c_k , p_k , and e are examples of how topology manifests itself. Results on ellipticity, Sobolev space estimates, and Fredholm theory are obtained from analysis.

Furthermore, the classification of topological phases, supersymmetric quantum mechanics, and the comprehension of anomalies in quantum field theory are all based on index theory in theoretical physics.

7.6 Summary

The Atiyah-Singer Index Theorem links analysis, geometry, and topology together. Special cases of the theorem yield fundamental results such as the Gauss-Bonnet, Riemann-Roch, and signature theorems. Interestingly, its general form has applications ranging from geometry to number theory and physics.

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